

Module 1, Lab 3 — Textbook Problem 1.2

Choosing an auto insurance provider

Name: Mac Mckay **Course:** Module 1 — What Is Data Science / Setting Up Your Toolkit

The problem. Three auto insurance providers, three customer ratings each (out of 10). Pick one and defend the choice.

```
In [1]: import pandas as pd
import matplotlib.pyplot as plt

df = pd.DataFrame({
    "provider": ["GEICO"] * 3 + ["Progressive"] * 3 + ["USAA"] * 3,
    "rating":    [4.7, 8.3, 9.2, 7.4, 6.7, 8.9, 3.8, 6.3, 8.1],
})

df
```

```
Out[1]:
```

	provider	rating
0	GEICO	4.7
1	GEICO	8.3
2	GEICO	9.2
3	Progressive	7.4
4	Progressive	6.7
5	Progressive	8.9
6	USAA	3.8
7	USAA	6.3
8	USAA	8.1

Min, mean, max

```
In [2]: summary = df.groupby("provider")["rating"].agg(["min", "mean", "max",
summary["range"] = summary["max"] - summary["min"]
summary.round(2)
```

Out[2]:

	min	mean	max	std	range
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provider					
GEICO	4.7	7.40	9.2	2.38	4.5
Progressive	6.7	7.67	8.9	1.12	2.2
USAA	3.8	6.07	8.1	2.16	4.3

Ranking by min, mean max

```
In [3]: ranks = pd.DataFrame({
    "by_mean": summary["mean"].rank(ascending=False).astype(int),
    "by_min": summary["min"].rank(ascending=False).astype(int),
    "by_max": summary["max"].rank(ascending=False).astype(int),
    "by_range": summary["range"].rank(ascending=True).astype(int), #
})
print("Rank 1 = best on that measure\n")
ranks
```

Rank 1 = best on that measure

Out[3]:

	by_mean	by_min	by_max	by_range
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provider				
GEICO	2	2	1	3
Progressive	1	1	2	1
USAA	3	3	3	2

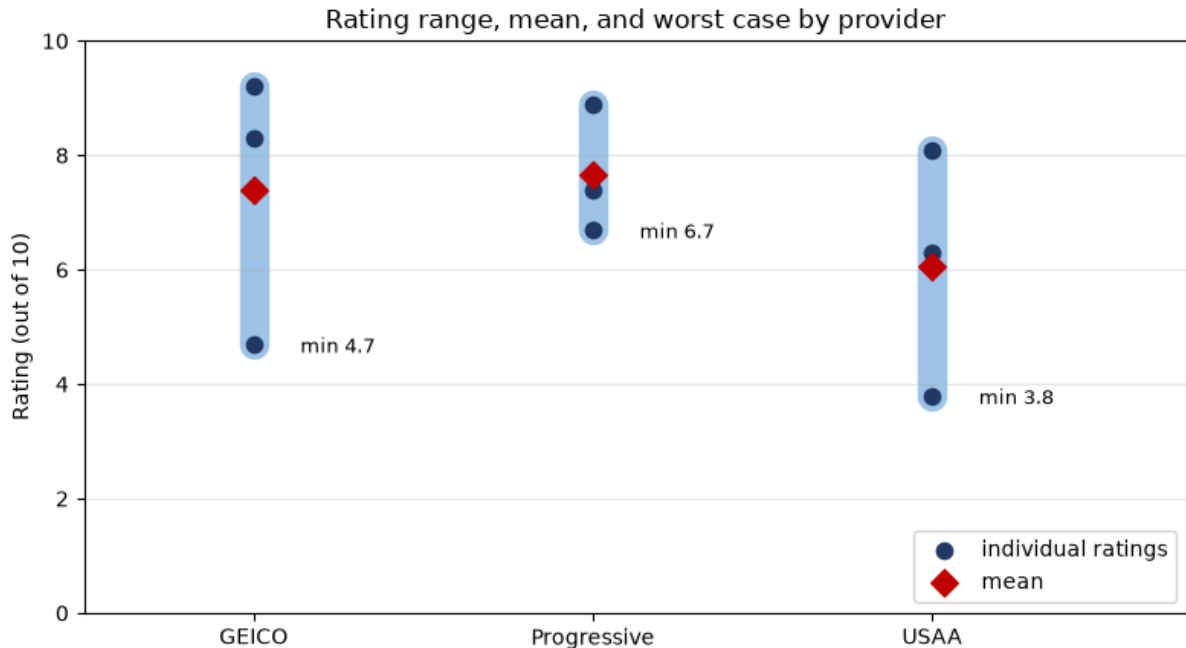
Progressive ranks first on mean, first on minimum, and first on consistency. The only measure it loses is `max` ; the single best review.

```
In [4]: fig, ax = plt.subplots(figsize=(8, 4.5))
providers = summary.index.tolist()

for i, p in enumerate(providers):
    lo, hi = summary.loc[p, "min"], summary.loc[p, "max"]
    ax.plot([i, i], [lo, hi], color="#9dc3e6", linewidth=14, solid_cap
    pts = df.loc[df["provider"] == p, "rating"]
    ax.scatter([i] * len(pts), pts, color="#1f3864", s=55, zorder=3, l
    ax.scatter([i], [summary.loc[p, "mean"]], color="#c00000", marker=
    ax.annotate(f"min {lo}", (i, lo), textcoords="offset points", xyte

ax.set_xlim(-0.5, len(providers) - 0.25)
ax.set_xticks(range(len(providers)))
ax.set_xticklabels(providers)
```

```
ax.set_ylabel("Rating (out of 10)")
ax.set_ylim(0, 10)
ax.set_title("Rating range, mean, and worst case by provider")
ax.grid(axis="y", alpha=0.3)
ax.legend(loc="lower right")
plt.tight_layout()
plt.show()
```



Written Answer

I would go with Progressive. Its average rating of 7.67 is the highest of the three, but the number that actually decides it for me is its minimum of 6.7, compared to 4.7 for GEICO and 3.8 for USAA. That minimum represents the worst experience a real customer reported, and for insurance specifically the worst case is what matters most, because you deal with your provider at the moment something has already gone wrong in your life. GEICO is the tempting pick here: it has the single highest rating in the table at 9.2 and an average only about a quarter point behind Progressive, so on a quick glance it looks competitive. But that average is being propped up by one excellent review while another customer rated it a 4.7, which means GEICO's quality is inconsistent and your experience depends heavily on luck. Progressive's three ratings all fall in a narrow 6.7 to 8.9 band, so its average is actually representative of what a typical customer gets rather than a blend of very good and very bad outcomes. USAA is out on both counts, with the lowest average and the lowest floor.

Summary table for reference

Provider	Min	Mean	Max	Range
GEICO	4.7	7.40	9.2	4.5
Progressive	6.7	7.67	8.9	2.2
USAA	3.8	6.07	8.1	4.3

In []: